Desion Making In C

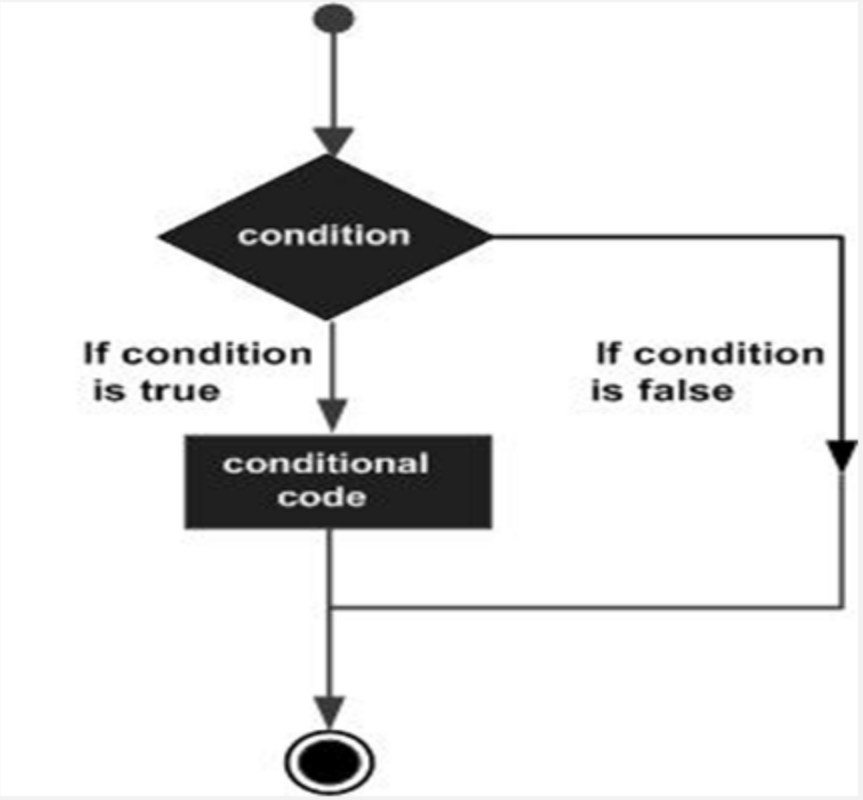
Objective Of Todays Session….

OBJECT™ TECHNOLOGIES

- Control Flow
- “C” Desion Making Statements
- The If Statement
- The If-else Statement
- Nested If
- Switch Case
- Conditional Operator (Ternary Operator)

Control Flow

Below is the general form of a typical decision making structure found in most of the programming languages



Desion Making Statements

OBJECT™ TECHNOLOGIES

1. if statement : An if statement consists of a Boolean expression followed by one or more statements.

2. if...else statement: An if statement can be followed by an optional else statement, which executes when the Boolean expression is false.

3. nested if statements :You can use one if or else if statement inside another if or else if statement(s).

4. if else ladder :you can use if else if statements.

5. switch statement : A switch statement allows a variable to be tested for equality against a list of values.

Desion Making Statements

6. Nested switch statements:
   You can use one switch statement inside another switch statement.

7. Ternary Operator:
   You can use Ternary Operator Exp1?Exp2:Exp3.

8. Nested Ternary operator:
   You can use one Ternary Operator inside another Ternary Operator.

www.object.co.in

Simple If Statement

- Simple if statement:
  The general form of a simple if statement is,

- Syntax:

```
if(expression)
{
   statement-inside;
}
   statement-outside;
```

  If the expression is true, then 'statement-inside' it will be executed, otherwise 'statement-inside' is skipped and only 'statement-outside' is executed.

Example

//Write a C Program which demonstrate use of if statement.

```
#include<stdio.h>
#include<conio.h>
void main()
{
   int x,y;
   x = 7;
   y = 5
   if(x > y)
   {
      printf("x is greater than y");
   }
   getch();
}
```

if else statement

- if...else statement

  The general form of a simple if...else statement is,

```
if(expression)
{
   statement-block1;
}
else
{
   statement-block2;
}
```

  If the 'expression' is true, the 'statement-block1' is executed, else 'statement-block2' is executed.

Example Of if- else statement

OBJECT™ TECHNOLOGIES

```
//Write a C Program which demonstrate use of if else-statement.
#include<stdio.h>
#include<conio.h>
void main()
{
   int x,y;
   x = 5;
   y = 7;
   if (x > y)
   {
      printf("x is greater than y");
   }
   else
   {
      printf("y is greater than x");
   }
   getch();
}
```

Nested If... else Statements

- Nested if....else statement:

  Syntax:

```
if(expression)
{
   if(expression1)
   {
      statement-block1;
   }
   else
   {
      statement-block2;
   }
}
else
{
   statement-block3;
}
```

if 'expression' is false the 'statement-block3' will be executed, otherwise it continues to perform the test for 'expression1' . If the 'expression1' is true the 'statement-block1' is executed otherwise 'statement-block2' is executed.

Use Of Nested If Statement

OBJECT™ TECHNOLOGIES

```
#include<stdio.h>
#include<conio.h>
void main()
{
   int a,b,c;
   clrscr();
   printf("Enter 3 number to compare: ");
   scanf("%d%d%d",&a,&b,&c);
   if(a > b)
   {
      if(a > c)
      {
         printf("a is greatest");
      }
      else
      {
         printf("c is greatest");
      }
```

```
}
else
   {
      if(b > c)
      {
         printf("b is greatest");
      }
      else
      {
         printf("c is greatest");
      }
   }
   getch();
}
```

if else if Ladder.

Syntax:

```
if(expression 1)
{
   statement-block1;
}
else if(expression 2)
{
   statement-block2;
}
else if(expression 3)
{
   statement-block3;
}
else
   default-statement;
```

The expression is tested from the top (of the ladder) to downwards. As soon as the true condition will found, the statement associated with it is executed.

Example of if else if Ladder.

```
//Write a C Program which demonstrate use of if-else-if ladder statement.
#include<stdio.h>
void main()
{
   int a;
   printf("Enter a Number: ");
   scanf("%d",&a);
   if(a > 0)
   {
      printf("Given Number is Positive");
   }
   else if(a == 0)
   {
      printf("Given Number is Zero");
   }
   else if(a < 0)
   {
      printf("Given Number is Negative");
   }
   getch();
}
```

Switch Statement .

- Switch statement is used to solve multiple option type problems for menu like program, where one value is associated with each option.
- The expression in switch case evaluates to return an integral value, which is then compared to the values in different cases, where it matches that block of code is executed, if there is no match, then default block is executed.

```
switch(expression)
{
  case ConstantValue  :
                        statement(s);
                        break;
  case ConstantValue  :
                        statement(s);
                        break;
  case ConstantValue  :
                        statement(s);
                        break;
  default : statement(s);
}
```

Switch Case

OBJECT™ TECHNOLOGIES

Rules for apply switch:

- With switch statement use only byte, short, int, char data type.
- You can use any number of case statements within a switch.
- Value for a case must be same as the variable in switch .
- Duplicate cases are not allowed in switch case.
- Limitations of switch: Logical operators and comparison operators cannot be used with switch statement.
  Example
  case k>=20: // is not allowed

Points To Remember

OBJECT™ TECHNOLOGIES

- We don't use those expressions to evaluate switch case, which may return floating point values or strings.

- It isn't necessary to use break after each block, but if you do not use it, all the consecutive block of codes will get executed after the matching block.

- Check Following Example, The output was supposed to be only A because only the first case matches, but as there is no break statement after the block, the next blocks are executed, until the cursor encounters a break.

- Default case can be placed anywhere in the switch case. Even if we don't include the default case switch statement Works

Switch Case

```
int i = 1;
switch(i)
{
   case 1:
      printf("A");       // No break
   case 2:
      printf("B");       // No break
   case 3:
      printf("C");
   break;
}
```

OBJECT™ TECHNOLOGIES

The ? : Operator

- It has the following general form Exp1 ? Exp2 : Exp3;

Where Exp1, Exp2, and Exp3 are expressions. Notice the use and placement of the colon.

The value of a ? expression is determined as below:

- Exp1 is evaluated. If it is true, then Exp2 is evaluated and becomes the value of the entire ? expression.
- If Exp1 is false, then Exp3 is evaluated and its value becomes the value of the expression

?: Operator

```
#include<stdio.h>
#include<conio.h>

void main()
{
    int a, b, c, large;
    clrscr();
    printf("Enter any three number: ");
    scanf("%d%d%d",&a,&b,&c);
    large=a>b ? (a>c?a:c) : (b>c?b:c);
    printf("Largest Number is: %d",large);
    getch();
}
```